



Impact of COVID-19 pandemic on cultural ecosystem services from urban green spaces: a case from English Bazar Urban Agglomeration, Eastern India

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Abstract

Urban green space (UGS) provides multiple cultural ecosystem services (CES) and enhances people's physical and mental well-being. The importance of UGS was more critical during crisis periods such as the COVID-19 pandemic. Recently, UGS has been under severe threats due to rapid urban expansion and climate change. Particularly, the cities in developing countries have experienced a dramatic loss of green spaces due to land use and land cover (LULC) change. This study aims to assess the impact of the COVID-19 pandemic on CES perception obtained from two public UGS, i.e., Subhankar Sishu Uddyan (SSU) and Bandh Road (BR) in English Bazar Urban Agglomeration (EBUA), Eastern India. A survey was performed to assess the valuation of CES, CES obtained from UGS, and the use of CES difference during partial (unlock phases) and after lockdown periods. The non-parametric tests were carried out to understand the significant differences between CES and UGS perceived by the respondents. Explanatory factor analysis was also performed to find the association among CES for each UGS. The results reported substantial variations of UGS use during partial and after lockdown periods. During the after-lockdown period, the UGS mainly were used for social relations (51%), spending time with relatives and friends (25%), spending time with children (11%) in SSU and walking (25%), spending time with relatives and friends (21%), and physical activities (17%) in BR. During the partial lockdown period, a significant difference between SSU and BR was recorded from social relations and health value among all CES. After the full lockdown period, a significant difference was observed in recreation, social relations, health, and aesthetic value. The frequency and duration of UGS use during partial lockdown periods substantially increased. About 12% of the total visitor visited UGS daily after the lockdown, whereas it was 4% during the partial lockdown. The study's findings will assist urban planners and policymakers in understanding the need for UGS to promote the well-being of the citizens.

Keywords Ecosystem services · Green spaces · COVID-19 · Well-being · Health value

Introduction

Urban green space (UGS) is highly vulnerable to rapid urban expansion and climate change (Bardhan et al. 2016; Xu et al. 2019). In the last few decades, there has been an increase in research focused on the UGS impact on urban population

well-being (e.g., Ma et al. 2019; Kothencz et al. 2017; Nath et al. 2018). Despite the UGS's importance, the UGS are under severe threat due to rapid urban expansion (Arshad et al. 2020; Wu et al. 2019), one of the major drivers of UGS loss (Girma et al. 2019). In developing countries such as India, unplanned urban expansion is considered a severe threat to UGS (Kaur et al. 2021). This decreases access to green spaces (Arshad et al. 2020) and the ecosystem services (ES) supplied. Due to this, several studies highlighted the importance of including UGS and ES in urban planning (Kabisch et al. 2015). Urban expansion in developing countries is growing dramatically. This process affects UGS tremendously in India, and ES is provided (Govindarajulu 2014; Sundaram 2011). Although urbanization poses severe threats to UGS, few studies in India focused on ES supplied

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by UGS (Ramaiah and Avtar 2019; Kadaverugu et al. 2021). More particularly, the cultural ecosystem services (CES) obtained from UGS remain unexplored in India.

UGS supplies habitats essential to prevent biodiversity loss in urban areas (Niemela 1999) that are key to supplying a vast number of regulating (e.g., air quality and climate regulation, noise reduction, carbon sequestration, water purification), provisioning (e.g., food, medicinal plants, biomass), and cultural (e.g., recreation, landscape aesthetics) ES vital to urban dwellers' well-being (Johnson et al. 2019; Giacchè et al. 2021; Pinto et al. 2021). Several psychological and physical health benefits can be obtained from UGS, such as reduced mental stress, depression, anxiety, physical fitness, and social interaction (Jabbar et al. 2021; Reyes-Riveros et al. 2021). Since establishing the Millennium Ecosystem Assessment (MEA, 2005), ES and their contributions to well-being have been assessed from diversified perspectives (e.g., quantitative and qualitative methods). CES quantification is challenging since CES perceptions and attitudes vary between cultures and must consider intangible values (Ko and Son 2018). Since the 1990s, urban CES research has become popular (Hubacek and Kronenberg 2013) and was mainly focused on recreation and tourism (Abildtrup et al. 2013; de Groot et al. 2010). Very little work has been conducted on other CES components such as spirituality, aesthetic value, and cognitive development since they are non-physical and challenging to quantify (MEA, 2005; Ko and Son 2018). Thus, UGS has become one of the critical measures of quality of life, development of the cities, and sustainable development (Zhu and Xu 2021; Claudia et al. 2022). Notably, the importance of UGS has been well recognized during crisis periods such as the COVID-19 pandemic (Claudia et al. 2022).

In December 2019, COVID-19 originated in Wuhan (China), spreading rapidly worldwide (Noszczyk et al. 2022). The World Health Organization (WHO) declared a global public health emergency on 11 March 2020. The heavily affected countries implemented lockdowns to restrict the spread of the disease (Lian and Feng 2022; Kumari and Toshniwal 2020). The public spaces, GS, were closed to prevent COVID-19 transmission (Benzell et al. 2020; Noszczyk et al. 2022; Gorzelany et al. 2022). In India, the lockdown was imposed on 25 March 2020, continuing until 31 May 2020. During this period, strict restrictions on access to public spaces (such as parks) were imposed. After 31 May 2020, unlock phases were implemented from 1 June 2020 and continued until 31 March 2022). During this un-lockdown phase, many relaxations were provided on access to public spaces. The COVID-19 pandemic outbreak substantially affected urban life and UGS usage (Venter et al. 2020). Human mobility was strongly restricted (Slater et al. 2020),

affecting people's physical and mental health. The limitations to obtaining CES from UGS were high (Richard and WCPO staff 2020; Reynolds and Forgione 2020). Access to public spaces was restricted (Rogers 2020; Yip and Chau 2020). Recent studies showed that UGS substantially reduced psychological stress (such as poor mental health and post-traumatic stress) that increased due to COVID-19 impacts (Geng et al. 2021; Slater et al. 2020). Therefore, examining the impact of the COVID-19 lockdown on CES supplied by UGS is essential. Thus, this study aims to examine the impact of COVID-19 amid the lockdown on CES supplied by UGS.

Several global studies recognized that the COVID-19 lockdown affected CES supplied by UGS (Ugolini et al. 2020; Mayen Huerta and Cafagna 2021; Berdejo-Espinola et al. 2021). Nevertheless, in the Indian context, most of the study was performed on various aspects of CES obtained from UGS, such as benefits from urban parks (Basu and Nagendra 2021), UGS with their necessity (Ramaiah and Avtar 2019), and UGS planning for climate changes mitigations (Govindarajulu 2014). However, to our knowledge, no studies were carried out to understand the impact of lockdown on the CES obtained from UGS. The outbreak of COVID-19 primarily affected the use, perception, and attitudes of the people (Grima et al. 2020; da Schio et al. 2021), and most of the study in India was performed on the impact of COVID-19 on air quality (Das et al. 2021; Naqvi et al. 2021; Sharma et al. 2020), water quality (Arif et al. 2021), urban vulnerability (Mishra et al. 2020), and urban slum vulnerability (Das et al. 2021), but the impact of COVID-19 pandemic on CES obtained from UGS has been ignored and unexplored. Thus, these gaps in the previous research studies inspired us to perform a study on the impact of COVID-19 amid the lockdown on CES obtained from the UGS. This study has immense scope to understand UGS's role in providing CES from a diversified perspective during the crisis periods. Thus, this research aims to assess the impact of the COVID-19 pandemic on the CES during partial and after-lockdown periods in English Bazar Urban Agglomeration (EBUA), Eastern India. The specific objectives of this work are (1) to understand the use of UGS during the partial lockdown and after the complete lockdown, (2) to compare the use of CES during the partial and complete lockdown, and (3) to find out the association CES during the partial and complete lockdown, respectively. Few research questions have been formulated to fulfill the objectives of the study. These research questions are as follows: (1) For what purposes were UGS used during partial and after-lockdown periods? (2) Is there any difference in CES usage between partial and after-lockdown periods? (3) To what extent are the CES associated?

Materials and methods

Study sites

Two UGS were selected to study COVID-19 impact on CES (Fig. 1). These UGS are located in the EBUA, Eastern India. EBUA is one of the largest urban agglomerations in West Bengal, with an area of 28 km² and a population of more than 0.3 million (Dutta and Das 2019). EBUA comprises two statutory towns (English Bazar Municipality and Old Malda Municipality) and three census towns (Bagbari, Chhatianmore, and Sahapur). Recently, EBUA experienced rapid urban growth, and it was considered the fastest-growing agglomeration in West Bengal (Dutta and Das 2019). The city experienced a rapid change in vegetation cover,

water bodies, and agricultural land due to dramatic land use and land cover change (Shaw and Das 2018; Dutta and Das 2019). The average temperature is 26 °C, and the average rainfall is around 1500 mm in EBUA (Dutta and Das 2019). There are very few UGS in English Bazar city, such as parks, playgrounds, and green spaces at the city and the neighbourhood level. These UGS are widely used for recreation, social relations, and physical activities. Some major UGS are SSU, BR, Brinda Bani Maidan, Malda College Ground, Prasanti Uddyan, Maldar Uthon, and Mahananda Park. Unfortunately, no large park or GS exist in the city. Among all the parks, SSU and BR are the most popular and widely used UGS and, therefore, selected for this work. SSU is located in the city's center and is a public park maintained by the English Bazar municipality. This park is not only used



Fig. 1 Location of the study sites: (a) India, (b) West Bengal, (c) EBUA, (i) SSU, and (ii) BR

by the people within the city, people outside the city visit the park, also SSU is mainly used for social relations and cultural activities. Band Road is one the most popular UGS in EBUA's southern part, along the Mahananda river. Urban people mainly use this green space for physical, social activities and recreational activities. The people use this UGS because of its location, particularly near the river and beauty.

CES studied and survey and data collection

In this study, six CES categories obtained from UGS were developed following the framework proposed by

the Millennium Ecosystem Assessment (MEA, 2005) and TEEB (2010). These six CES were categorized into 24 sub-CES based on previous works (Barrena et al. 2014; Norton et al. 2012; Bieling et al. 2014) (Supplementary questionnaire S1; Section. 1 to 5). The CES studied are recreation, social relations, inspirations, aesthetic value, sense of place, and health value. The details of the CES are presented in Table 1.

Previously, to conduct the questionnaires, we tried to understand how much the visitors value the CES obtained from SSU and BR. The visitors were asked to identify the six most relevant.

Table 1 CES used in this study

CES	Description	Reference	Subcategories
Recreations	UGS provide recreational facilities outdoors, such as walking and relaxing	Plieninger et al. (2013), Bieling (2014)	UGS provides sufficient recreational facilities [REC1] UGS are good for recreational activities [REC2] UGS contains natural beauty for recreation [REC3] UGS are attractive [REC4]
Social relations	UGS provide opportunities to build social cohesion and bonding through walking and gossiping. It offers opportunities to form new social relations with neighbors and unknown people	Plieninger et al. (2013), Brown (2012)	UGS provides opportunities for social relations [SR5] UGSs provide opportunities to interact with neighbors/friends [SR6] UGS provide opportunities for new social relations [SR7] UGS provide opportunities to build social cohesion [SR8]
Inspiration	UGS stimulate new ideas, thoughts, and creative expression	Plieninger et al. (2013), Burkhard et al. (2012)	UGS are ideal for relaxation [IN9] UGS are a source of inspiration [IN10] UGS are essential for enhancing creativity [IN11] UGS decreases mental stress [IN12]
Aesthetic value	UGS are the source of aesthetic value due to their pristine nature and facilitating opportunities for various aesthetic pleasures of ecosystems such as natural beauty, landscape design, and visual satisfaction	Plieninger et al. (2013), Bieling (2014)	UGS attracts me for its landscape beauty [AV13] UGS attracts me for its beautiful design [AV14] UGS provides opportunities to experience nature [AV15] UGS is embraced by lush greenery [AV16]
Sense of place	The UGS foster a sense of attachment to the places	Bieling (2014)	I miss this UGS while moving outside the city [SP17] UGS remind my past memories [SP18] I am emotionally attached to this UGS [SP19] UGS are comfortable and relaxing [SP20]
Health value	UGS provide opportunities for regular and continuous activities that enhance people's physical and mental health	Ko and Son (2018)	UGS reduces mental stress [HV21] UGS provide opportunities for physical fitness [HV22] UGS provides mental refreshment [HV23] UGS enhance the quality of life [HV24]

Face-to-face interviews were performed in SSU and in BR ($N = 215$). The visitors were randomly selected from the UGS to evaluate the impact of COVID-19 amid the lockdown on the use, attitudes, and perception of CES in EBUA. Individual and focus group discussions (comprising five to six people) with the park visitors were conducted as well. In total, 104 individuals were interviewed in SSU and 111 in BR. The survey was performed in mid of July to 29 August 2021 for partial lockdown (unlock phases). During the survey, the respondents were also asked about CES use, attitudes, and perception during the after-lockdown periods. In SSU, the survey was conducted one time, i.e., from 4 to 6 PM, and in the case of BR, it was carried out during the morning (6 AM to 8 AM) and the evening (4 PM to 7 PM), respectively. The survey was conducted during this time because it is in this time that visitor frequency is high. In the case of BR, this GS is mostly used for different purposes in the morning (mainly for physical activities) and evenings (for multiple reasons), such as walking and spending time with friends. The questionnaires were developed in English. However, then questions were asked in Bengali Dialect. The UGS has also assessed the frequency of UGS visits and duration spent in UGS. The details of the methodology are presented in Fig. 2.

The respondents valued the six CES and the different sub-categories using a 5-point Likert scale ranging from very important/highly agree (5) to very low important/highly disagree (1). The survey was focused on the following aspects (Supplementary S2):

1. Socio-demographic attributes of the visitors
2. Perceived valuation of CES from UGS
3. Activities performed at UGS during partial and after lockdown periods
4. Duration and frequency of UGS use during partial and after-lockdown periods
5. Perception toward CES (subcategories), respectively

Statistical analysis

In this study, several descriptive statistics (median, quartiles) were carried out. The median refers to the middle most value of the datasets meaning 50% of datasets are smaller or equal to the median, and 50% are higher or higher than the median. The first and third quartiles refer to the 25% and 75% of the datasets arranged in increasing order. In this study, the median and the first and third quartiles have been calculated based on the CES obtained from the UGS. There were variations in the use of UGS during partial and after

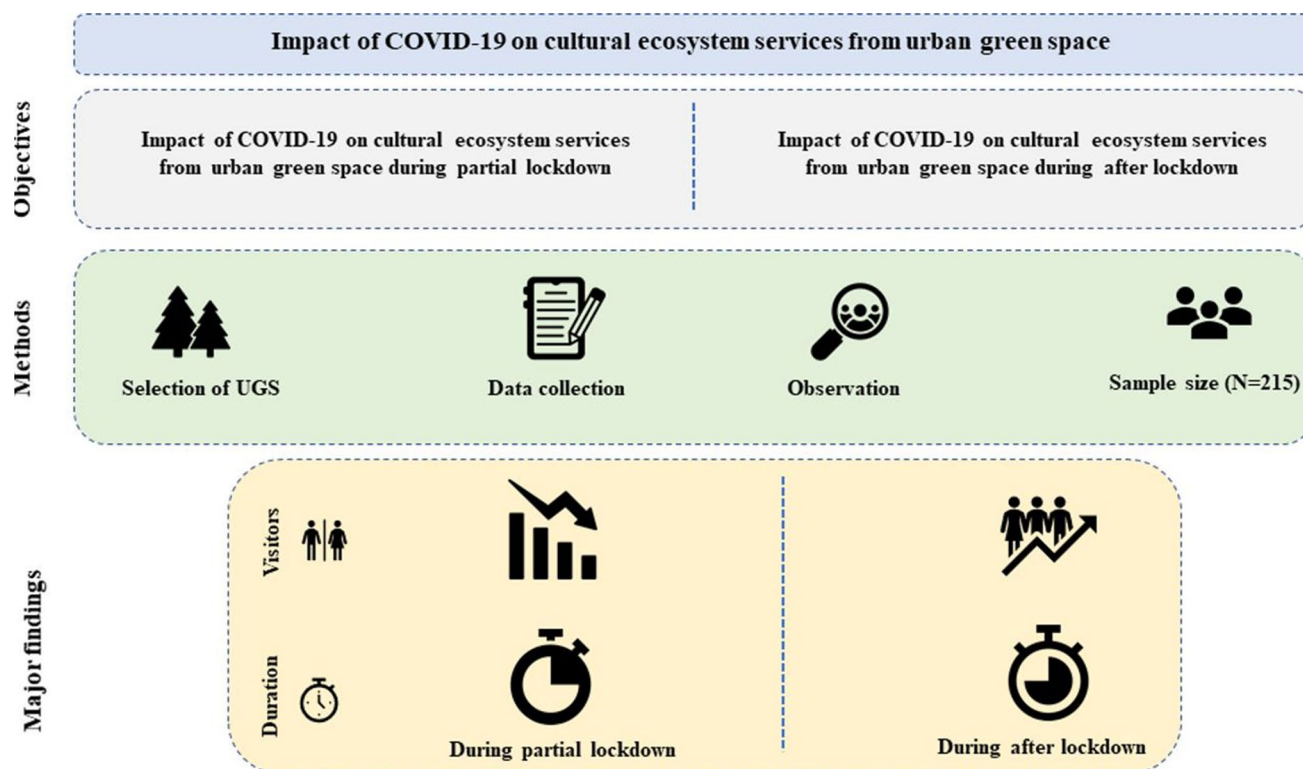


Fig. 2 Framework of the entire research work

lockdown. Therefore, it is important to understand the difference between partial and after-lockdown periods. Thus, a Mann–Whitney (MW) *U* test was applied to find differences in the CES perception between study sites and partial and after a lockdown in SSU and BR. Significant differences were also considered at a *p* value < 0.05. A factor analysis (FA) was performed to determine the association between CES for each UGS during partial and after lockdown periods. FA was performed based on the correlation matrix. All the statistical analysis was carried out with SPSS 22. Figures were created using online plotly (<https://chart-studio.plotly.com/>).

Results

UGS visitors' profile

This study surveyed 55% of males and 45% of females from SSU and 61% of males and 39% of females from BR. The details socio-demographic profile of the respondents is presented in Table 2. In SSU, the majority of the visitors were 20–30 years old (76%), followed by 30–40 years old

(13%), 40–50 years old (9%), and above 50 years old (2%). On the other hand, in BR, most of the visitors belonged to the age group of 20–30 years (58%), followed by 30–40 years (19%), 40 to 50 years (11%), and above 50 years (12%). The majority of the respondents had high school (class 8 to 12) (32%) level followed by Bachelor (30.5%), elementary (from class 1 to 7) (23.5%), and above master (13.5%). The educational status of the visitors was different in the two UGS. In SSU, 62% of visitors were students, 8% worked as government employees, 14% were business people, and 16% were domestic. In BR, 49% of visitors were students, 21% were housewives, 19.0% were business people, and 11.0% worked as government employees (Supplementary Table S1).

Except for education level, recreation was significantly higher in BR for gender, education level, and occupation. Regarding social relations, except for age, all the remaining variables were significantly higher in SSU than in BR. Inspiration and aesthetic value were significantly higher in BR than in SSU in all the variables. Sense of place was significantly higher in BR than in SSU for gender and occupation. Finally, the health value was significantly higher in BR than in SSU in all the studied variables (Table 2).

Table 2 Difference in socio-demographic characteristics of the visitor toward CES

CES	UGS	Gender ^a	Age ^b	Educational level ^c	Occupation ^d
Recreation (REC)	SSU	4.2 (3.97, 4.27)	2.49 (2.31, 2.84)	3.20 (2.97, 3.51)	3.14 (2.88, 3.37)
	BR	4.6 (4.44, 4.71)	4.48 (4.38, 4.52)	3.16 (2.93, 3.23)	4.51 (4.35, 4.66)
	MW <i>p</i> value	*	**	ns	**
Social relation (SR)	SSU	3.83 (3.56, 3.99)	2.40 (2.20, 2.60)	3.43 (3.13, 3.70)	2.24 (2.16, 2.44)
	BR	2.91 (2.65, 3.22)	2.48 (2.08, 2.64)	1.37 (1.08, 1.87)	1.96 (1.71, 2.17)
	MW <i>p</i> value	**	ns	**	**
Inspiration (IN)	SSU	1.9 (1.77, 2.15)	2.35 (2.24, 2.79)	2.41 (2.08, 2.68)	2.60 (2.44, 2.95)
	BR	3.67 (3.43, 4.17)	3.78 (3.44, 4.10)	3.59 (3.40, 3.76)	3.42 (3.19, 3.68)
	MW <i>p</i> value	**	**	**	**
Aesthetic value (AV)	SSU	3.05 (2.8, 3.33)	2.7 (2.55, 2.9)	2.90 (2.65, 3.17)	3.07 (2.81, 3.27)
	BR	4.80 (4.69, 4.88)	4.51 (4.28, 4.63)	3.45 (3.25, 3.66)	4.27 (4.17, 4.38)
	MW <i>p</i> value	**	**	**	**
Sense of place (SP)	SSU	2.86 (2.69, 2.97)	2.94 (2.68, 3.09)	2.36 (2.25, 2.66)	1.96 (1.79, 2.25)
	BR	3.86 (3.58, 4.01)	3.19 (2.95, 3.38)	2.04 (1.76, 2.43)	3.01 (2.36, 3.30)
	MW <i>p</i> value	**	ns	ns	**
Health value (HV)	SSU	1.26 (1.10, 1.49)	1.46 (1.26, 1.66)	1.70 (1.59, 2.04)	2.26 (2.01, 2.52)
	BR	4.82 (4.69, 4.87)	4.14 (3.96, 4.22)	3.34 (3.02, 3.53)	3.87 (3.75, 4.03)
	MW <i>p</i> value	**	*	**	**

The median (1st quartile, 3rd quartile) values are based on a Likert scale ranging from 1 (low importance) to 5 (very high importance). Different letters represent significant differences at **p* < 0.05, ***p* < 0.01, and ****p* < 0.001. ns, non-significant differences at *p* < 0.05

SSU Subhankar Sishu Uddyan, BR Bandh Road

^a1: Male, 2: Female

^b1: 20–30, 2: 30–40, 3: 40–50, 4: > 50

^c1: Elementary, 2: High School, 3: Bachelor, 4: above the Master

^d1: Students, 2: Government Employees, 3: Businessmen, 4: Housewives

Reasons and duration of visits to UGS

During the partial lockdown, SSU was mostly used for social relations (37%), followed by spending time with relatives and friends (21%), experiencing nature and its aesthetic beauty (15%), and mental refreshment and relaxation (13%), respectively (Fig. 3). On the other hand, during a partial lockdown, BR was mostly used for spending time with relatives and friends (29%), followed by walking (21%), physical activities (14%), fresh environment (10%), and mental refreshment and relaxation (8%). During the partial lockdown, about 43% of visitors visited UGS less than 1 time in a month, followed by once a week (22.5%) and 1 to 4 times a month, respectively. Most of the visitors (32%) spent up to 1 h, followed by 1 to 2 h (30%), and 2 to 3 h (22%) during the partial lockdown.

During after lockdown periods, SSU was mostly used for social relations (51%), followed by spending time with relatives and friends (25%) and spending time with children (11%). In BR, UGS was mainly used for walking (25%),

followed by spending time with relatives and friends (21%), physical activities (17%), and experiencing nature and its natural beauty (12%). Thus, from the overall result, it was reported that UGS was mostly used for social relations (27%), followed by spending time with relatives and friends (23%), walking (14%), and physical activities (9%) (Supplementary Figs. S1 and S2). After the lockdown, most of the visitors visited UGS less than 1 time in a month (32%), followed by 1 to 4 times a month (21.5%) and 2 to 4 times a week (1%). However, most of the visitors visited BR 2 to 4 times a week (28%), followed by less than 1 time in a month (23%) and daily (19%). After the lockdown, 47% of visitors spent 1 to 2 h, followed by up to 1 h (34%) and 2 to 3 h (12%).

Comparison difference during partial and after lockdown toward CES

Regarding CES's subcategories, there was a significant difference in CES between the two UGS. In recreation, except

Fig. 3 Use of UGS with different activities in SSU: **a** children playing in the park, **b** social relations, **c** gathering of visitors in the park, **d** view of the park and in BR, **e** gathering of people in the evening, **f** people spending time with their friends, **g** physical activities in the morning, and **h** riverside view of the green space



REC1 (“UGS provides sufficient recreational facilities”), a significant difference was observed for all the recreational CES (REC2, REC3, and REC4). For social relations, a significant difference was reported from SR5 (“UGS provides opportunities for social relation”), SR7 (“UGS provides opportunities to new social relations”), SR8 (“UGS provides opportunities to build social cohesion”), and SR6 (“UGS provides opportunities to interact with neighbors/friends”). Among inspiration CES, a significant difference during partial lockdown was found for IN10 (“UGS are a source of inspirations”) and IN11 (“UGS are important for enhancing creativity”). Sense of place had the only significant difference for SP20 (“UGS are comfortable and relaxing”). Regarding aesthetic value, a significant difference was reported between AV13 (“UGS attracts me for its landscape beauty”) and AV15 (“UGS are embraced by lush greenery”). There were also significant differences between CES within SSU and BR during the partial lockdown period (Table 3).

There was also a significant variation of CES values between SSU and BR after lockdown periods. In the case of recreation and social relations, a significant difference was recorded from all the CES. Among all the CES under inspirations, a significant difference was found for IN9 (“UGS are ideal for relaxation”), IN11 (“UGS are important for enhancing creativity”) and IN12 (“UGS decreases mental stress”). There was also a significant difference in aesthetic value between SSU and BR for AV13 (“UGS attracts me for its landscape beauty”), AV15 (“UGS provides opportunities to experience nature”) ($p > 0.05$), and AV16 (“UGS are embraced by lush greenery”). In the case of sense of place, there was a significant difference between SSU and BR during after lockdown periods for SP17 (“I miss this UGS while moving outside the city”), SP18 (“UGS reminds my past memories”), and SP19 (“I am emotionally attached with this UGS”).

Lastly, in the case of health, a significant difference was recorded from HV21 (“UGS reduces mental stresses”), HV22 (“UGS provides opportunities for physical fitness”) and HV23 (“UGS provides mental refreshment”), respectively. The Tukey post hoc test showed significant differences among CES within SSU and BR (Table 4). Significant differences between SSU and BR were found because BR was provided more CES during partial and after lockdown.

Multivariate analysis

Factor analysis (FA) was performed for each UGS during partial and after-lockdown periods. The exploratory FA loading for two UGS is detailed in Supplementary Tables S2 and S3. During the partial lockdown, factor 1 was explained by social relations (SR5), aesthetic value (AV16), and sense of place (SP17 and SP20). In contrast, factor 2 was formed by recreation (REC3), inspirations (IN9 and IN10), aesthetic

Table 3 Visitors’ perception according to CES subcategory obtained during the partial lockdown (median (1st quartile, 3rd quartile))

CES	ID	SSU Median (1st quartile, 3rd quartile)	BR Median (1st quartile, 3rd quartile)	MW <i>p</i> value
Recreation	REC1	2.5 (2.3, 2.6)	2.8 (2.4, 3.1)	ns
	REC2	1.6 (1.6, 2.0)	3.6 (3.5, 4.0)	*
	REC3	2.0 (1.7, 2.1)	2.7 (2.6, 2.8)	*
	REC4	2.6 (2.5, 2.7)	2.9 (2.6, 3.1)	ns
Social relations	SR5	3.2 (3.0, 3.4)	2.4 (2.2, 2.6)	*
	SR6	2.3 (1.8, 2.6)	1.9 (1.4, 2.6)	ns
	SR7	2.8 (2.6, 2.9)	2.1 (1.9, 2.2)	*
	SR8	2.1 (1.9, 2.5)	1.4 (1.3, 1.6)	*
Inspiration	IN9	2.3 (1.6, 2.5)	2.7 (2.3, 3.1)	ns
	IN10	1.6 (1.3, 1.6)	2.3 (2.1, 2.5)	*
	IN11	1.7 (1.6, 2.0)	2.4 (2.0, 2.6)	*
	IN12	2.1 (2.0, 2.3)	2.8 (1.8, 2.9)	ns
Aesthetic value	AV13	1.6 (1.6, 2.0)	2.8 (2.7, 2.9)	*
	AV14	1.8 (1.7, 2.1)	2.5 (1.8, 2.6)	ns
	AV15	1.3 (1.2, 1.5)	2.6 (2.3, 2.9)	*
	AV16	1.7 (1.6, 2.0)	2.5 (2.1, 2.6)	ns
Sense of place	SP17	1.9 (1.5, 2.2)	2.30 (1.8, 2.6)	ns
	SP18	1.6 (1.4, 2.0)	2.10 (1.9, 2.2)	ns
	SP19	2.2 (1.9, 2.6)	2.20 (2.1, 2.5)	ns
	SP20	1.9 (1.4, 2.1)	2.60 (2.5, 3.0)	*
Health value	HV21	2.20 (1.7, 2.9)	2.90 (2.6, 3.0)	ns
	HV22	1.2 (1.1, 1.3)	2.80 (2.6, 2.9)	*
	HV23	1.4 (1.3, 2.1)	2.50 (2.0, 2.9)	*
	HV24	1.8 (1.7, 2.0)	2.60 (2.5, 2.8)	*

Values are based on a Likert scale ranging from 1 (disagree) to 5 (strongly agree). *Significant differences at $p < 0.05$. ns, non-significant differences at $p < 0.05$. The details of the abbreviations are presented in the Supplementary section (S2)

SSU SubhankarSishuUddyan, BR Bandh Road

value (AV14), and sense of place (SP19) for SSU. Although six groups can be identified, the CES parameters were scattered in this case. The first group, CES for SSU, includes SR5, SR7 (social relations), and REC4 (recreations). Another group was formed with IN9 (inspiration) and HV21 (health value). The third group was composed of SR8 (social relation), IN12 (inspiration), and REC1 (recreation), and the fourth group was formed by SR6 (social relations) and SP19 (sense of place) (Fig. 4).

In the case of BR, factor 1 was explained by social relations (SR5, SR6, and SR8), inspirations (IN11 and IN12), and health value (HV23), respectively. Factor 2 was formed by social relations (SR6 and SR7), inspirations (IN10 and IN11), aesthetic value (AV14 and AV15), and health value (HV22). In the case of BR, three groups can be identified. SP20, IN9, REC4, and REC18 formed the first group. This

Table 4 Visitors' perception according to CES subcategory obtained after the lockdown (median (1st quartile, 3rd quartile))

CES	ID	SSU Median (1st quartile, 3rd quartile)	BR Median (1st quartile, 3rd quartile)	MW <i>p</i> value
Recreation	REC1	3.8 (3.5, 4.0)	4.82 (4.71, 4.85)	*
	REC2	1.6 (1.5, 2.0)	4.87 (4.84, 4.93)	*
	REC3	3.2 (2.7, 3.5)	4.59 (4.41, 4.64)	*
	REC4	3.6 (3.2, 3.8)	4.59 (4.36, 4.71)	*
Social relations	SR5	4.6 (4.5, 4.7)	2.76 (2.57, 3.15)	*
	SR6	4.0 (4.0, 4.2)	2.94 (2.68, 3.21)	*
	SR7	4.2 (4.1, 4.3)	2.00 (1.70, 2.15)	*
	SR8	4.2 (4.0, 4.3)	2.75 (2.65, 3.13)	*
Inspiration	IN9	2.8 (2.5, 3.1)	4.03 (3.85, 4.25)	*
	IN10	3.0 (2.6, 3.5)	3.75 (3.65, 3.93)	ns
	IN11	2.5 (1.9, 3.0)	4.24 (4.18, 4.26)	*
	IN12	2.9 (2.7, 3.3)	4.27 (4.25, 4.56)	*
Aesthetic value	AV13	2.8 (2.7, 3.2)	4.24 (4.21, 4.31)	*
	AV14	3.1 (2.7, 3.3)	4.10 (3.76, 4.47)	ns
	AV15	2.2 (2.0, 2.6)	4.42 (4.24, 4.55)	*
	AV16	2.7 (2.6, 2.8)	3.48 (3.27, 3.60)	ns
Sense of place	SP17	3.2 (3.0, 3.5)	4.06 (3.65, 4.33)	*
	SP18	3.5 (3.0, 3.5)	4.29 (3.96, 4.49)	*
	SP19	3.3 (3.0, 3.5)	4.32 (4.26, 4.49)	*
	SP20	3.6 (3.5, 3.7)	4.06 (3.73, 4.27)	ns
Health value	HV21	2.6 (2.3, 2.8)	4.60 (4.54, 4.67)	*
	HV22	1.8 (1.6, 2.1)	4.76 (4.68, 4.84)	*
	HV23	3.6 (3.3, 3.9)	4.36 (4.25, 4.47)	*
	HV24	3.4 (2.7, 3.6)	3.60 (3.54, 4.11)	ns

Values are based on a Likert scale ranging from 1 (disagree) to 5 (strongly agree). *Significant differences at $p < 0.05$. ns, non-significant differences at $p < 0.05$. The details of the abbreviations are presented in the Supplementary section (S2)

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group was inversely related to AV16, SP17, SP19, and AV14. Finally, the third group was formed by HV22, REC1, and HV21.

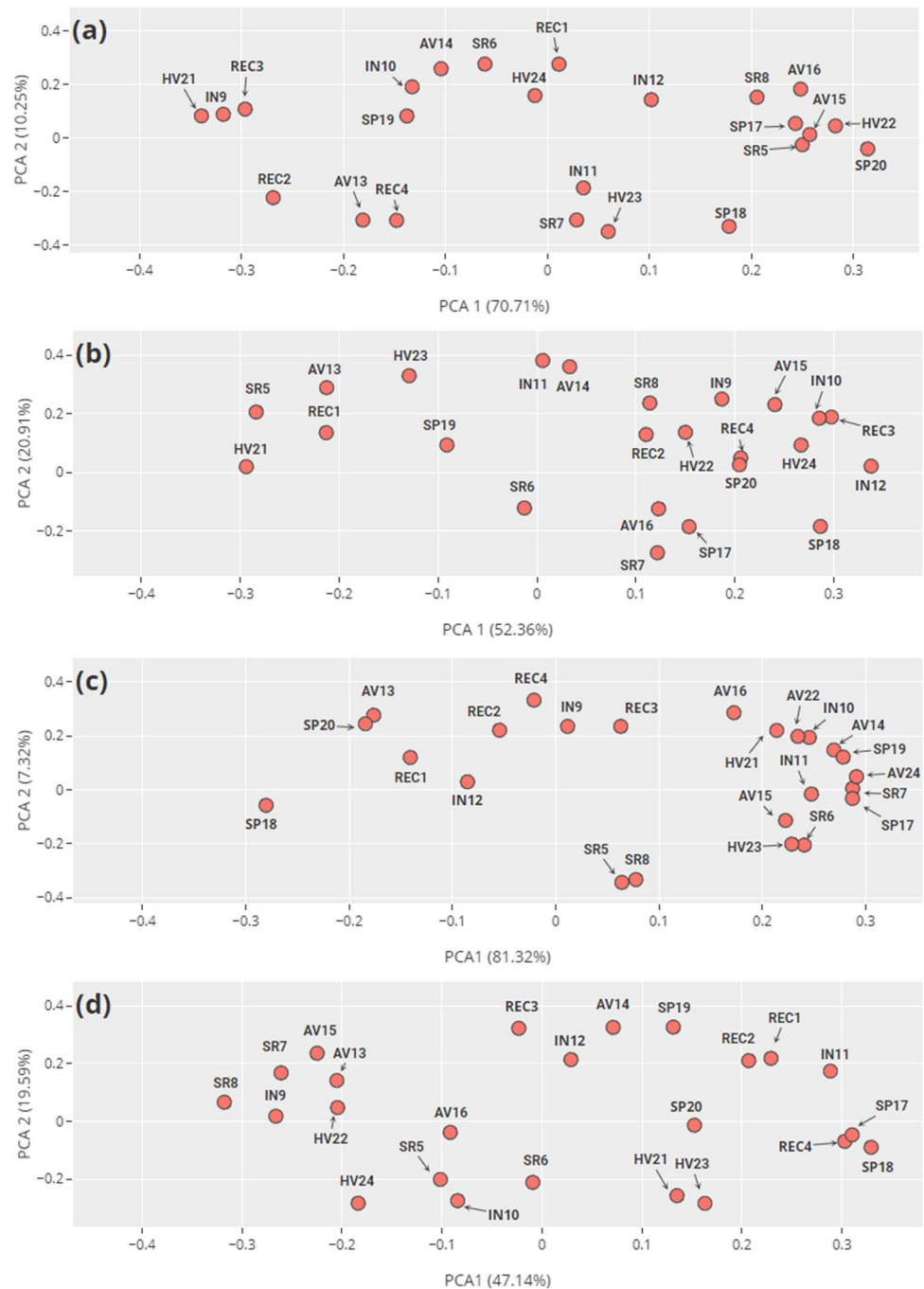
During after-lockdown periods, in the case of SSU, factor 1 was formed by recreation (REC3), social relations (SR8), inspirations (IN10, IN11, and IN12), aesthetic value (AV15), and health value (HV24). Factor 2 was explained by social relations (SR5), inspirations (IN11), aesthetic value (AV13 and AV14), and health value (HV21 and HV24), respectively. In the case of SSU, the CES services are scattered distributed, and six groups were identified. The first group was formed by SR5, SR6, SR7, and SR8. This group is inversely related to REC2, HV22, and AV15. Finally, the third group was composed of three CES parameters with distinctive negative values.

In the case of BR, factor 1 was formed by recreations (REC4), inspirations (IN11), sense of place (SP17 and SP18), and health value (HV21). The factor was explained by recreation (REC3), inspiration (IN11 and IN12), and sense of place (SP19), respectively. In the case of SSU, three groups of CES can be identified. AV14 and SP17 related to aesthetic and health values formed the first group. A second group was formed by social relations (SR5, SR6, and SR8), respectively. A third group was composed of diversified CES such as health value (HV23, HV21, HV22), recreations (REC1, REC2, REC3, REC4), sense of place (SP19, SP20), and inspiration (IN11, IN9, IN12).

Discussion and conclusions

The study showed that COVID-19 amid lockdown substantially impacted CES obtained from two UGS in an Indian urban agglomeration (EBUA), Eastern India. Valuation of CES obtained from the UGS, frequency of use, and duration varies between two UGS. There were substantial variations between the two UGS toward the valuation of CES during partial and after lockdown periods. Both the UGS reported that the most commonly mentioned CES was recreation, followed by social relations, aesthetic value, and inspirations, respectively. During after-lockdown period, relatively higher importance was given to social relations (4.88), followed by recreation (4.56), aesthetic value (4.23), and inspiration (4.12). In SSU and from BR, relatively higher importance was given to aesthetic value (4.79), followed by health (4.77), inspiration (4.58), recreation (4.32), and social relations (3.68). In the case of the use of UGS, it was reported that UGS was mostly used for social relations (27%), followed by spending time with relatives and friends (23%), walking (14%), and physical activities (9%). During after lockdown period, BR was mainly used for walking (25%), followed by spending time with relatives and friends (21%), physical activities (17%), and experiencing nature and its

Fig. 4 Relation of PCA factor 1 and factor 2. **a** During partial and **b** after the lockdown period from SSU and **c** during partial and **d** after the lockdown period from BR. The details of the abbreviations are presented in the Supplementary section (S2)



natural beauty (12%). The frequency and duration of spending time in UGS increased after lockdown periods compared to partial lockdown periods. For example, after lockdown periods, about 31% of the total respondents visited UGS 1 to 4 times (16% during partial lockdown periods) in SSU. Similarly, 28% of the respondents in BR visited UGS 2 to 4 times weekly (19% during partial lockdown periods). As per the perception toward the subcategories of CES, it was observed that about 70 to 60% of the total visitors reported that BR provides sufficient recreational facilities, and these UGS are

good for walking and physical exercise, respectively. Only 40% of visitors were satisfied with the recreational services provided by SSS. The visitors were less satisfied with the recreation due to mainly being more diminutive in size and gathering of people. Thus, from the study, it was well recognized that CES obtained from the UGS plays a significant role, and visitors depend highly on UGS for various CES.

From the results of the study, it was well documented that there were differences in CES between UGS over the socio-demographic attributes in EBUA. This finding was

similar to other studies (Pinto et al. 2021). For the UGS, age, gender, and educational status had significant differences. Previous works also identified this (Wen et al. 2020; Riechers and Barkmann 2018; Sun et al. 2019). The findings of the study suggest that planning strategies must be implemented to enhance accessibility to meet social equity (Pinto et al. 2021). The results showed that about 40% of respondents visited BR due to mainly physical activities, and 45% visited BR to enjoy the environment (such as fresh environment, esthetic beauty, and mental refreshment) after the lockdown. Thus, it helped to improve physical as well as mental health. In the previous literature, it was well documented that UGS play a significant role in enhancing mental and physical health (Lee et al. 2015; Astell-Burt and Feng 2019; Ugolini et al. 2020; Enssle and Kabisch 2020). According to Annerstedt et al. (2012), UGS promote physical activities, helps recover illness faster, enhances general mental health, and reduces stress and mental disorders. Mainly, in densely populated areas like EBUA, the contributions of UGS are more crucial (Bateman et al. 2013; Keeler et al. 2019). The survey result showed that users substantially increased after lockdown periods due mainly to physical activities and enhancing mental health. In many recent studies, it was documented that the physical and mental health of the people were affected mainly by UGS during the COVID-19 pandemic (Xie et al., 2020; Ugolini et al. 2020). Thus, the multi-functional activities of UGS largely affect the use of UGS (Kaczynski and PotwarkaLR 2008; Zwierzchowska et al. 2018).

From the study's findings, it was well documented that UGS are used for multiple perspectives, and the use of CES from the UGS substantially increased after lockdown periods. Now, it is important to compare the findings of the study with other previous studies. Recently, several studies were carried out to assess the impact of the COVID-19 pandemic on the attitudes, use, and perception of UGS worldwide (Berdejo-Espinola et al. 2021; Geary et al. 2021; Pouso et al. 2021). For example, Beckmann-Wübbelt et al. (2021) performed a study in southwestern Germany. They showed that visits to the peri-urban forest (PUF) increased during the COVID-19 pandemic, and PUFs were mostly used for enhancing mental health (such as stress reduction). As per the study of Ugolini et al. (2021) in Italy, it was reported that UGS were important for relaxation and physical exercises during the pandemic periods. Ugolini et al. (2020) also studied multiple countries (such as Croatia, Israel, Spain, Italy, Lithuania, and Slovenia). Their findings revealed that UGSs were mostly used for physical activities, relaxation, and experiencing nature, respectively. Behavioral changes during the COVID-19 pandemic were also reported, such as walking in tree-lined streets (Spain) and small urban gardens (Italy) and traveling outside the city to green areas (Lithuania), respectively. Geng et al. (2021) studied the impact of

the COVID-19 pandemic on global urban park visits. Their findings reported that the number of park visitors increased before 16 February 2020 (before the extreme outbreak of COVID-19) compared to prior to the COVID-19 pandemic periods. From the result of the study, it was reported that the frequency and duration spent in UGS increased after lockdown periods due to the relaxation of the restrictions. The frequency of visits to UGS increased by 30% after the lockdown compared to the partial lockdown period. During the partial lockdown period (in July and August 2020), there was a strict restriction on the use and accessibility of public space use, and there were limited opportunities to visit UGS in the city.

Policy implications

The COVID-19 pandemic allowed realizing the importance of UGS for the well-being of the people. It also revealed that the presences of UGS (like urban parks) are crucial to enhancing the city dwellers' quality of life, particularly during crisis periods like COVID-19 amid lockdown. In many previous studies, it was well documented that the frequency of UGS uses substantially increased during the COVID-19 pandemic (Lopez et al. 2020; Beria and Lunkar 2021). The role of UGS in enhancing physical and mental health during COVID-19 was well recognized (Lin et al. 2019; Schipperijn et al. 2010). It denotes the importance of UGS during crisis periods, and these UGS were very important for the mental and physical well-being of urban dwellers (Venter et al. 2020; Kleinschroth and Kowarik 2020; Grima et al. 2020). In urban areas, city dwellers remained locked at home and spent less time outside. People became disconnected from social interactions and felt lonely. The urban dwellers became mentally depressed, and there were minimal opportunities for outdoor activities due to the lockdown. Thus, importance and dependency substantially increased after the lockdown periods of the COVID-19 pandemic (Ugolini et al. 2020; Noszczyk et al. 2022). Thus, from the study's findings, UGS (parks, gardens, patches of green spaces, and tree-lined streets) must be promoted to enhance urban health and well-being. In developing countries like India, conservation and management of UGS have become severe challenges due to rapid unplanned and haphazard urban expansions (Ramaiah and Avtar 2019; Dhyani et al. 2018). In the Indian context, it was well reported that there was a dramatic loss of green (vegetation cover) cover due to rapid urban expansions and conversion of green cover (vegetation cover) to other land use, respectively (Nagendra et al. 2012; Bhat et al. 2017). UGS planning and management is one of the great challenges to city planners and policymakers in fast-growing, developing cities like India. Previous studies exposed concerns regarding the loss of urban green in cities due to rapid urban expansion and

the deterioration of environmental quality in Indian cities (Sundaram 2011; Govindarajulu 2014). In EBUA, several studies were performed by Dutta and Das (2019), Shaw and Das (2018), and Pal and Ziaul (2017), and their findings reported that there was a dramatic increase in the urban area and rapid transformation of land use and land cover. According to the study of Dutta and Das (2019), from 1991 to 2016, 17% of vegetation cover was transformed into agricultural land, and 2.33% was converted into a built-up area. Therefore, urgent actions must be adopted to sustain urban environmental quality across India. In the Indian context, the roles of UGS were well documented in previous work to promote the urban environment, mitigate urban heat island (UHI) effects, enhance mental health well-being, and obtain CES from UGS (Chaturvedi et al. 2013; Rahman et al. 2021; Mukherjee et al. 2017). The status of UGS (at city and neighborhood scale) and per-capita availability of UGS across the cities of India were assessed to achieve urban sustainability and promote urban health well-being (Anguluri and Narayanan 2017; Singh 2018; Shekhar and Aryal 2019). Therefore, effective urban landscape planning is urgently required to promote urban green space management at the city and neighborhood levels.

This study has immense scope to understand UGS's role in providing CES and its role in crisis periods. This study is crucial to capture CES's diverse perspective during and after the partial lockdown periods. Despite this, a few limitations of the study can be reported. First, in this study, July–August 2020 and July–August 2021 were considered partial (unlock phases) and after-lockdown periods. Second, during the pandemic periods, an online survey was performed due to a lack of equipment, and minimal responses from the survey were collected to maintain COVID-19 protocols onsite. Thus, a sample for the study may not be sufficient. Third, in this study, the urban green space sites were selected based on the frequency of use and importance of UGS sites. Future researchers must consider the size of UGS and diverse sources of ES to understand visitors' perceptions of ES better.

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Author contribution MD, AD, and SS conceptualized and designed the study. MD and SS collected and analyzed the data with the support of AD and PP, and wrote the paper. MD and SS interpreted the analysis. MD, AD, and PP read and modified the paper. All four have contributed extensively for data interpretation and discussions.

Data availability The datasets used during the current study are available from the corresponding author upon reasonable request.

Declarations

Consent to participate All participants provided their consent for participating in the interview.

Conflict of interest The authors declare no competing interests.

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